

| Question | Marking Guidance                                                                                                                                                                                                                                                          | Mark | Comments                                                                                                                        |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|---------------------------------------------------------------------------------------------------------------------------------|
| 02.1     | <p>The bacteriophage has a capsid and the bacterium has a cell-surface membrane;</p> <p>Third box down</p>                                                                               | 1    | <p>Reject if more than one box with tick. Ignore crossed-out ticks</p> <p>Accept tick to right or left of correct box</p>       |
| 02.2     | <p>Correct number of times between 13.0/12.96 <b>and</b> 13.9/13.92 scores 2 marks;;</p> <p>One mark if correct sizes in ranges of 150.7nm to 154.4nm/ 0.151µm to 0.154µm <b>and</b> 1953.5nm to 2097.6nm/ 1.954µm to 2.098µm;</p>                                        | 2    | <p><b>Both</b> lengths required for 1 mark credit</p> <p>Accept refs to 150/0.15 and 2000/2</p> <p>Ignore number of sig fig</p> |
| 02.3     | <p>The bacteriophage makes no difference to the number of (living) <u>bacteria</u>/there will be no difference in the number of (live) <u>bacteria</u> in treated and untreated mice/ there will be no difference in the number of (live) <u>bacteria</u> in A and B;</p> | 1    | <p>Ignore no difference between mice</p>                                                                                        |

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|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 02.4 | <p>1. Log scale (on graph) shows big range in number of bacteria</p> <p>OR</p> <p>Use of suitable data from log scale to give the range in number of bacteria;</p> <p>2. Some samples too many to count (so dilute) but some countable (so don't dilute)</p> <p><b>OR</b></p> <p>Use of figures from graph relating to ease (or otherwise) of counting</p> <p>Example. 631 000 000 bacteria would be too big to count (without serial dilution), 100 000 bacteria is small enough to count;</p> | 2 | <p>1 and 2 <b>Do not</b> accept simple statements of <math>\log_{10}</math> values from graph</p> <p>Look for answers in standard form</p> <p><b>Group A</b></p> <p>Mean between 79 million and 100 million</p> <p>Range 4 million to 631 million</p> <p><b>Group B</b></p> <p>Mean between 100 thousand and 126 thousand</p> <p>Range 3.98/4 up to 251 thousand</p> |
| 02.5 | <p>1. (Bacteriophage) reduces number of bacteria;</p> <p>2. (In all cases/mice because) ranges don't overlap;</p> <p>3. But big range of effect/some mice a big reduction/a few mice with big falls in number</p> <p>OR</p> <p>Doesn't bring bacteria down to 0 in any/works for some (mice) but not for all;</p>                                                                                                                                                                               | 3 | <p>Do not accept just quotes of <math>\log_{10}</math> figures direct from graph</p> <p>Ignore refs to significance</p> <p>2. Reject ref to SD / SE</p>                                                                                                                                                                                                              |

